

Probability and Random Processes

Markov chains. Exercises

1. A ticket sales system utilises two computers: one operates actively, while the other remains idle and is designated to assume operations if the active computer fails. The probability that the active computer fails during a given day is $0 < p < 1/2$. If a failure occurs, the affected computer becomes operative again only on the second day following the failure.

Let (x, y) represent the state of the system, where x denotes the number of computers ready for operation and y indicates the number of days a computer has been under repair.

- Model the system using a Markov chain. Classify all possible states and determine the corresponding transition probability matrix.
- Assuming both computers are operative today, identify the most probable system state two days later.
- Compute the stationary distribution. In the stationary regime, determine the probability that neither computer is operative on a given day when $p = 0.3$.

Solution

- Consider a Markov chain with states (i, j) , where $i \in \{0, 1, 2\}$ represents the current number of operative computers and $j \in \{0, 1\}$ indicates the number of remaining days until at least one of the computers becomes operative. Therefore, the state space is $\{S_1, S_2, S_3, S_4\}$, where:
 - $S_1 = (2, 0)$: the two computers are operative.
 - $S_2 = (1, 0)$: the computer in service has failed today, and the other is operative.
 - $S_3 = (1, 1)$: one computer failed yesterday, and the other is operative.
 - $S_4 = (0, 1)$: one computer failed yesterday, and the other has failed today.

The transition probability matrix is:

$$\mathbf{P} = \begin{pmatrix} q & p & 0 & 0 \\ 0 & 0 & q & p \\ q & p & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

- If today the two computers are operative, the probability distribution after two days is:

$$(1, 0, 0, 0)\mathbf{P}^2 = (1, 0, 0, 0) \begin{pmatrix} q^2 & pq & pq & p^2 \\ q^2 & pq & p & 0 \\ q^2 & pq & pq & p^2 \\ q & p & 0 & 0 \end{pmatrix} = (q^2, pq, pq, p^2).$$

Since $p < 1/2$, the most probable state after two days is S_1 , that is, the two computers are operative.

- If $\boldsymbol{\pi} = (\pi_1, \pi_2, \pi_3, \pi_4)$ is the stationary distribution vector, then the equation $\boldsymbol{\pi} = \boldsymbol{\pi}\mathbf{P}$ and the normalization condition gives:

$$\begin{aligned} \pi_1 &= q\pi_1 + q\pi_3 \\ \pi_2 &= p\pi_1 + p\pi_3 \\ \pi_3 &= q\pi_2 + \pi_4 \\ \pi_4 &= p\pi_2 \\ 1 &= \pi_1 + \pi_2 + \pi_3 + \pi_4 \end{aligned}$$

The solution is:

$$\pi_1 = \frac{q}{q + 2p + p^2}, \quad \pi_2 = \pi_3 = \frac{p}{q + 2p + p^2}, \quad \pi_4 = \frac{p^2}{q + 2p + p^2}.$$

Hence, the probability that the two computers are not operative if $p = 0.3$ is

$$\pi_4 = \frac{p^2}{q + 2p + p^2} = \frac{p^2}{1 + p + p^2} \approx 0.06475.$$

2. An elevator operates by moving up and down, stopping at each floor of the building. It has a maximum capacity of three passengers. At each floor, one of the following events can occur: either one passenger gets in, or one passenger gets off, or no one gets on or off. The probability of one passenger boarding the elevator is $1/4$ when the elevator is not full, and it is zero when the elevator is at full capacity. Conversely, the probability of one passenger getting off is $1/4$ when the elevator is not empty, and it is zero when the elevator is empty.

(a) Determine the Markov chain, classify its states and compute its transition probability matrix.

If there are currently i passengers in the elevator,

- (b) What is the most probable number of passengers after two stops?
 (c) In the long term, what is the average number of passengers?
 (d) If T_i denotes the time (number of stops) that the elevator is full for the first time, what is the expected value of T_i ?

Solution

- (a) Let X_n denote the number of passengers in the elevator at the n -th stop. The value of X_n , given X_{n-1} , does not depend on any previous values X_k for $k < n-1$. Therefore, the sequence $\{X_n : n \geq 0\}$ forms a Markov chain, which set of states is $S = \{0, 1, 2, 3\}$. The transition probability matrix is:

$$P = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & \frac{1}{4} & \frac{3}{4} \end{pmatrix}$$

- (b) The probability distribution after two stops is given by

$$\mathbf{p} \cdot P^2 = \mathbf{p} \begin{pmatrix} \frac{10}{16} & \frac{5}{16} & \frac{1}{16} & 0 \\ \frac{5}{16} & \frac{6}{16} & \frac{4}{16} & \frac{1}{16} \\ \frac{1}{16} & \frac{4}{16} & \frac{6}{16} & \frac{5}{16} \\ 0 & \frac{1}{16} & \frac{5}{16} & \frac{10}{16} \end{pmatrix},$$

where $\mathbf{p}$ is the current probability distribution vector. When there are i passengers, $\mathbf{p}$ will have zeros in all positions except for a 1 in the i -th position. This setup indicates that $\mathbf{p} \cdot P^2$ corresponds to the i -th row of the matrix P^2 . Since the maximum value in this i -th row occurs at the i -th position, the most likely number of passengers after two stops remains the same as it was at the current stop.

- (c) The chain is aperiodic (each state has a non-zero probability of remaining invariant), and all states are intercommunicating. By the ergodic theorem, the chain has a stationary probability vector $\boldsymbol{\pi} = (\pi_0, \pi_1, \pi_2, \pi_3)$, which represents the limiting distribution, to be solved from the equations

$$\begin{aligned} \boldsymbol{\pi} P &= \boldsymbol{\pi} \\ \sum_i \pi_i &= 1, \end{aligned}$$

which gives the solution $\boldsymbol{\pi} = (1/4, 1/4, 1/4, 1/4)$. Therefore, over time, we can expect an average of $\sum_{i=0}^3 i \cdot \pi_i = \left(\sum_{i=0}^3 i\right)/4 = 1.5$ passengers.

- (d) Clearly, $\mathbb{E}(T_3) = 0$ (the elevator is already full). On the other hand, for the other three variables, comparing their expectations with those of the next stop, we have:

$$\mathbb{E}(T_0) = \frac{3}{4} \mathbb{E}(T_0) + \frac{1}{4} \mathbb{E}(T_1) + 1$$

$$\mathbb{E}(T_1) = \frac{1}{4} \mathbb{E}(T_0) + \frac{1}{2} \mathbb{E}(T_1) + \frac{1}{4} \mathbb{E}(T_2) + 1$$

$$\mathbb{E}(T_2) = \frac{1}{4} \mathbb{E}(T_1) + \frac{1}{2} \mathbb{E}(T_2) + \frac{1}{4} \mathbb{E}(T_3) + 1$$

The solution to this system is $\mathbb{E}(T_0) = 24$, $\mathbb{E}(T_1) = 20$, and $\mathbb{E}(T_2) = 12$.

3. Consider a system consisting of four balls, two white and two black. Two balls are placed in urn A and the remaining two in urn B . The state of the system is defined by the number of white balls in urn A . The following process is repeatedly applied: a ball is selected at random from urn A and transferred to urn B , followed by the random selection of a ball from urn B to be transferred to urn A .
- (a) Determine the transition probability matrix $\mathbf{P}$ for the associated Markov chain.
 - (b) Identify the stationary distribution. Assess whether this stationary distribution also serves as a limiting distribution and provide justification. If applicable, compute $\lim_{n \rightarrow \infty} \mathbf{P}^n$.
 - (c) Suppose two balls are randomly selected from the four. Calculate the probabilities of drawing 0, 1, or 2 white balls. Relate these results to the findings from the previous question.

Solution

- (a) Let X_n be the number of white balls in urn A after n transformations. We have

$$\mathbb{P}(X_n = 0 \mid X_{n-1} = 0) = \frac{1}{3}, \quad \mathbb{P}(X_n = 1 \mid X_{n-1} = 0) = \frac{2}{3}, \quad \mathbb{P}(X_n = 2 \mid X_{n-1} = 0) = 0.$$

$$\mathbb{P}(X_n = 0 \mid X_{n-1} = 1) = \frac{1}{6}, \quad \mathbb{P}(X_n = 1 \mid X_{n-1} = 1) = \frac{2}{3}, \quad \mathbb{P}(X_n = 2 \mid X_{n-1} = 1) = \frac{1}{6}.$$

$$\mathbb{P}(X_n = 0 \mid X_{n-1} = 2) = 0, \quad \mathbb{P}(X_n = 1 \mid X_{n-1} = 2) = \frac{2}{3}, \quad \mathbb{P}(X_n = 2 \mid X_{n-1} = 2) = \frac{1}{3}.$$

Thus $\{X_n : n \geq 0\}$ is a Markov chain with transition probability matrix

$$\mathbf{P} = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} & 0 \\ \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ 0 & \frac{2}{3} & \frac{1}{3} \end{pmatrix}.$$

- (b) The chain is irreducible. The balance equations are

$$\frac{2}{3}\pi_0 = \frac{1}{6}\pi_1$$

$$\frac{1}{3}\pi_1 = \frac{2}{3}\pi_0 + \frac{2}{3}\pi_2$$

$$\frac{2}{3}\pi_2 = \frac{1}{6}\pi_1$$

By symmetry, we have that $\pi_0 = \pi_2$. From the second equation, we get that $\pi_1 = 4\pi_0$. Hence, the stationary probability distribution vector is

$$\boldsymbol{\pi} = \left(\frac{1}{6}, \frac{4}{6}, \frac{1}{6} \right).$$

The chain is aperiodic (we have loops in all the states). Therefore, π is also the limiting distribution and

$$\lim_{n \rightarrow \infty} \mathbf{P}^n = \begin{pmatrix} \frac{1}{6} & \frac{4}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{4}{6} & \frac{1}{6} \\ \frac{1}{6} & \frac{4}{6} & \frac{1}{6} \end{pmatrix}.$$

(c) Let N be the number of white balls when selecting two balls. We have

$$\mathbb{P}(N=0) = \frac{\binom{2}{0}\binom{2}{2}}{\binom{4}{2}} = \frac{1}{6}, \quad \mathbb{P}(N=1) = \frac{\binom{2}{1}\binom{2}{1}}{\binom{4}{2}} = \frac{4}{6}, \quad \mathbb{P}(N=2) = \frac{\binom{2}{2}\binom{2}{0}}{\binom{4}{2}} = \frac{1}{6}.$$

These probabilities coincide with the limiting probabilities because, after many transformations, the effect is to shuffle the balls completely randomly.

4. Consider a particle performing a random walk on a directed graph with vertex set $\{1, 2, 3, 4\}$ and adjacency matrix $\mathbf{A} = (a_{ij})$:

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

Here, $a_{ij} = 1$ if and only if there is a directed edge from node i to node j ; otherwise, $a_{ij} = 0$. If the particle is located at node i , it randomly selects one of its adjacent neighbors (with equal probability) and moves to that node.

- Analyse the random walk using a Markov chain framework. Identify the type of Markov chain represented.
- Determine the stationary distribution. Assess whether it also serves as a limiting distribution and provide justification. For each node i , $1 \leq i \leq 4$, compute the expected number of steps required for the particle to return to i .
- Suppose the random walk begins at node i , where $3 \leq i \leq 4$. Calculate the expected number of steps until the particle first visits either node 1 or node 2. Solve this problem using two approaches: first-step analysis and by converting the Markov chain into an absorbing chain.

Solution

- The nodes of the directed graph correspond to the states of a Markov chain with transition probability matrix

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 1/3 & 1/3 & 1/3 & 0 \end{pmatrix}$$

It is a finite irreducible Markov chain. Thus, all the states are non-null recurrent.

- The balance equations are:

$$\begin{aligned} \pi_1 &= \frac{1}{2}\pi_2 + \frac{1}{3}\pi_4 \\ \pi_2 &= \pi_1 + \frac{1}{2}\pi_3 + \frac{1}{3}\pi_4 \\ \pi_3 &= \frac{1}{2}\pi_2 + \frac{1}{3}\pi_4 \\ \pi_4 &= \frac{1}{2}\pi_3 \end{aligned}$$

The solution of this system that satisfies $\sum_{i=1}^4 \pi_i = 1$ is the stationary distribution:

$$\boldsymbol{\pi} = \left(\frac{6}{25}, \frac{10}{25}, \frac{6}{25}, \frac{3}{25} \right).$$

The chain is aperiodic. (For instance, starting at node 1, it is possible to return to 1 in two or three steps. Thus, the period of state is 1; hence, the period of all the states is $d = 1$.) Therefore, the stationary distribution $\boldsymbol{\pi}$ is also the limiting distribution of the chain.

The expected number of steps τ_i since the particle starts at i until it returns to the same node i is the mean recurrence time of state i . Hence,

$$\tau_1 = \frac{1}{\pi_1} = \frac{25}{6}, \quad \tau_2 = \frac{1}{\pi_2} = \frac{5}{2}, \quad \tau_3 = \frac{1}{\pi_3} = \frac{25}{6}, \quad \tau_4 = \frac{1}{\pi_4} = \frac{25}{3}.$$

- (c) Let t_i be the expected number of steps until the particle hits either node 1 or node 2 for the first time, given that the random walk starts at i , $i = 3, 4$. We have

$$\begin{aligned} t_3 &= \frac{1}{2} + \frac{1}{2}(1 + t_4) = 1 + \frac{1}{2}t_4 \\ t_4 &= \frac{1}{3} + \frac{1}{3} + \frac{1}{3}(1 + t_3) = 1 + \frac{1}{3}t_3 \end{aligned}$$

The solution is $t_3 = 9/5$ and $t_4 = 8/5$.

The question can also be solved by converting states 1 and 2 into absorbing states. The fundamental matrix of the new absorbing chain is $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$, where $\mathbf{Q}$ is the transition probability matrix between the transient states 3 and 4:

$$\mathbf{Q} = \begin{pmatrix} 0 & 1/2 \\ 1/3 & 0 \end{pmatrix}.$$

Thus,

$$\mathbf{N} = \begin{pmatrix} 6/5 & 3/5 \\ 2/5 & 6/5 \end{pmatrix}.$$

The expected times to absorption are

$$t_3 = \frac{6}{5} + \frac{3}{5} = \frac{9}{5}, \quad t_4 = \frac{2}{5} + \frac{6}{5} = \frac{8}{5}.$$

5. Consider two urns containing a total of four balls. At each step, one of the four balls is randomly selected and transferred from its current urn to the other urn. Define the states as the number of balls in the first urn.

- Provide justification that the process constitutes a Markov chain and determine the corresponding transition probability matrix.
- Determine the stationary distribution. Assess whether this stationary distribution also serves as a limiting distribution and provide justification for your answer.
- Assume initially that the first urn contains all four balls. Calculate the expected number of movements required until it contains all four balls again.
- Now, suppose that each urn initially contains two balls. Calculate the expected number of movements required until all four balls are located in a single urn.

Solution

- (a) Let X_n represent the number of balls in the first urn after n movements (steps). The sequence $\{X_n : n \geq 0\}$ forms a Markov chain characterized by the following transition probability matrix:

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1/4 & 0 & 3/4 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 3/4 & 0 & 1/4 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

(b) The chain is irreducible. The balance equations are

$$\begin{aligned}\pi_0 &= \frac{1}{4}\pi_1 \\ \pi_1 &= \pi_0 + \frac{1}{2}\pi_2 \\ \pi_2 &= \frac{3}{4}\pi_1 + \frac{3}{4}\pi_3 \\ \pi_3 &= \frac{1}{2}\pi_2 + \pi_4 \\ \pi_4 &= \frac{1}{4}\pi_3\end{aligned}$$

By symmetry, we have $\pi_0 = \pi_4$ and $\pi_1 = \pi_3$. Moreover, from the above equations, $\pi_1 = 4\pi_0$ and $\pi_2 = 2(\pi_1 - \pi_0) = 6\pi_0$. Hence, the stationary probability distribution vector is

$$\boldsymbol{\pi} = \left(\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16} \right).$$

The chain is periodic with a period of 2, which means that it does not possess a limiting distribution.

- (c) If the first urn initially contains four balls, the expected number of movements until it contains all the balls again is the mean recurrence time of state 4, that is, $1/\pi_4 = 16$.
- (d) We can solve the question by converting 0 and 4 into absorbing states. The fundamental matrix of the absorbing Markov chain obtained in this way is $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$, where

$$\mathbf{Q} = \begin{pmatrix} 0 & 3/4 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 3/4 & 0 \end{pmatrix}$$

is the transition probability matrix of the transient states 1, 2 and 3. Therefore

$$\mathbf{N} = \begin{pmatrix} 5/2 & 3 & 3/2 \\ 2 & 4 & 2 \\ 3/2 & 3 & 5/2 \end{pmatrix}.$$

If initially each urn contains two balls, the expected number of movements until all the balls are in the same urn is given by the sum of the elements of the second row of $\mathbf{N}$, that is 8.

We can also solve the question by a first-step analysis. Let t_i be the expected time until state 0 or state 4 is visited for the first time, given that the chain starts at i , $i = 1, 2, 3$. By symmetry, we have $t_1 = t_3$. Moreover,

$$t_2 = \frac{1}{2}(t_1 + 1) + \frac{1}{2}(t_3 + 1) = \frac{1}{2}t_1 + \frac{1}{2}t_3 + 1 = t_1 + 1,$$

$$t_1 = \frac{1}{4} + \frac{3}{4}(t_2 + 1) = \frac{3}{4}t_2 + 1.$$

Solving these two equations, we obtain $t_2 = 8$ again.

6. Analyse an absorbing Markov chain with state space $S = \{1, 2, 3, 4\}$, given the following transition probabilities.

$$p_{11} = 1, \quad p_{21} = \frac{1}{2}, \quad p_{22} = 0, \quad p_{23} = \frac{1}{4}, \quad p_{33} = 1, \quad p_{41} = \frac{1}{2}, \quad p_{42} = \frac{1}{2}.$$

- (a) Classify all states and present the transition probability matrix in canonical form.
- (b) Calculate the fundamental matrix, then use it to determine the absorption probabilities and the expected times to absorption.
- (c) Recompute these parameters using a first-step analysis.

Solution

- (a) Since $p_{11} = 1$ and $p_{33} = 1$, states 1 and 3 are absorbing, and, hence, states 2 and 4 are transient. We have $p_{24} = 1 - p_{21} - p_{22} - p_{23} = 1/4$ and $p_{43} = p_{44} = 0$ because $p_{41} + p_{42} = 1$. Reordering the state space as $\{2, 4, 1, 3\}$ the transition probability matrix is in canonical form:

$$P = \begin{pmatrix} 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix},$$

where

$$Q = \begin{pmatrix} 0 & \frac{1}{4} \\ \frac{1}{2} & 0 \end{pmatrix}, \quad R = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 0 \end{pmatrix}.$$

- (b) The fundamental matrix is

$$N = (I - Q)^{-1} = \begin{pmatrix} 1 & -\frac{1}{4} \\ -\frac{1}{2} & 1 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{8}{7} & \frac{2}{7} \\ \frac{4}{7} & \frac{8}{7} \end{pmatrix}.$$

Hence, the absorption probabilities are given by

$$B = NR = \begin{pmatrix} \frac{8}{7} & \frac{2}{7} \\ \frac{4}{7} & \frac{8}{7} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 0 \end{pmatrix} = \begin{pmatrix} \frac{5}{7} & \frac{2}{7} \\ \frac{6}{7} & \frac{1}{7} \end{pmatrix},$$

that is

$$b_{21} = \frac{5}{7}, \quad b_{23} = \frac{2}{7}, \quad b_{41} = \frac{6}{7}, \quad b_{43} = \frac{1}{7}.$$

Moreover, the expected times to absorption are

$$\tau_2 = \frac{8}{7} + \frac{2}{7} = \frac{10}{7}, \quad \tau_4 = \frac{4}{7} + \frac{8}{7} = \frac{12}{7}.$$

- (c) We have

$$\begin{aligned} \tau_2 &= \frac{1}{2} + \frac{1}{4} + \frac{1}{4}(\tau_4 + 1) = 1 + \frac{1}{4}\tau_4 \\ \tau_4 &= \frac{1}{2} + \frac{1}{2}(\tau_2 + 1) = 1 + \frac{1}{2}\tau_2 \end{aligned}$$

The solution of this system is $\tau_2 = 10/7$ and $\tau_4 = 12/7$.

Analogously,

$$\begin{aligned} b_{21} &= \frac{1}{2} + \frac{1}{4}b_{41} \\ b_{41} &= \frac{1}{2} + \frac{1}{2}b_{21} \end{aligned}$$

Hence, $b_{21} = 5/7$ and $b_{41} = 6/7$, and thus, $b_{23} = 1 - b_{21} = 2/7$ and $b_{43} = 1 - b_{41} = 1/7$.

7. Consider a binary sequence of length four, denoted as $x_1x_2x_3x_4$, where each x_i belongs to $\{0, 1\}$. At each step, one of the four bits x_i is selected uniformly at random, and its value is flipped with probability $1/2$. Let X_n represent the number of ones in the sequence after n steps.

- (a) Demonstrate that if $X_n = i$ for $0 \leq i \leq 4$, then the probability that $X_{n+1} = i + 1$ is $(4 - i)/8$. Determine the transition probability matrix for the Markov chain $\{X_n : n \geq 0\}$. Assess whether the chain is irreducible and aperiodic, providing justification for each conclusion.

- (b) Find the stationary distribution and demonstrate that $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k) = \binom{4}{k}/16$. Provide an explanation for this result.
- (c) Assuming the initial sequence is 0000, apply first-step analysis to determine the expected number of steps required to obtain a sequence containing two ones and two zeros. Additionally, calculate the expected number of steps if the initial sequence is 1000 or 0010.
- (d) Re-examine the previous question by modelling the process as an appropriate absorbing Markov chain.

Solution

- (a) Suppose that after n steps the sequence contains i ones. At time $n + 1$ the sequence will have $i + 1$ ones if the bit chosen at random is a zero and it is changed to one, the sequence will have $i - 1$ ones if the chosen bit is a one at it changes to zero, and the sequence will have the same number of ones, i , if the selected bit does not change its value. Hence, for $0 \leq i \leq 4$,

$$\mathbb{P}(X_{n+1} = k \mid X_n = i) = \begin{cases} \frac{4-i}{4} \cdot \frac{1}{2}, & k = i + 1 \\ \frac{i}{4} \cdot \frac{1}{2}, & k = i - 1 \\ \frac{1}{2}, & k = i \\ 0, & \text{otherwise} \end{cases}$$

Therefore, the transition probability matrix is

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{8} & \frac{1}{2} & \frac{3}{8} & 0 & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 \\ 0 & 0 & \frac{3}{8} & \frac{1}{2} & \frac{1}{8} \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

The directed graph representing the chain is strongly connected (there is a directed path between any pair of nodes) and it has loops. Thus, the Markov chain is irreducible and aperiodic.

- (b) The balance equations are

$$\begin{cases} \frac{1}{2}\pi_0 = \frac{1}{8}\pi_1 \\ \frac{1}{2}\pi_1 = \frac{1}{2}\pi_0 + \frac{1}{4}\pi_2 \\ \frac{1}{2}\pi_2 = \frac{3}{8}\pi_1 + \frac{3}{8}\pi_3 \\ \frac{1}{2}\pi_3 = \frac{1}{4}\pi_2 + \frac{1}{8}\pi_4 \\ \frac{1}{2}\pi_4 = \frac{1}{8}\pi_3 \end{cases} \quad (1)$$

Considering the symmetries, we deduce that $\pi_0 = \pi_4$ and $\pi_1 = \pi_3$. The solution of the system (1) satisfying $\sum_{k=0}^4 \pi_k = 1$ is

$$\pi = \left(\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16} \right).$$

Notice that $\pi_k = \binom{4}{k}/16$.

Since the chain is aperiodic, the limiting distribution exists and is given by the stationary probabilities:

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k) = \pi_k = \frac{\binom{4}{k}}{16}, \quad k = 0, 1, 2, 3, 4.$$

In the long run, each bit in the sequence has a probability of $1/2$ of being 0 or 1. Thus, the probability that the sequence contains k ones is given by $\binom{4}{k}/16$.

- (c) When the sequence contains two ones and two zeros, the chain is visiting state 2. Let τ_i be the expected number of steps until the chain visits for the first time state 2, starting from state i . Since the starting sequence is 0000, we want to calculate τ_0 .

Performing a first-step analysis we have

$$\begin{cases} \tau_0 &= \frac{1}{2}(\tau_0 + 1) + \frac{1}{2}(\tau_1 + 1) &= 1 + \frac{1}{2}\tau_0 + \frac{1}{2}\tau_1 \\ \tau_1 &= \frac{1}{8}(\tau_0 + 1) + \frac{1}{2}(\tau_1 + 1) + \frac{3}{8} &= 1 + \frac{1}{8}\tau_0 + \frac{1}{2}\tau_1 \end{cases} \quad (2)$$

The solution of (2) is $\tau_0 = 16/3$ and $\tau_1 = 10/3$.

Either if the starting sequence is 1000 or 0010, the initial state of the Markov chain is state 1. Therefore, the expected number of steps until the chain hits state 2 is τ_1 .

- (d) Let us consider the absorbing Markov chain formed by states 0, 1 and the absorbing state 2, which transition probability matrix is

$$\mathbf{P}' = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{8} & \frac{1}{2} & \frac{3}{8} \\ 0 & 0 & 1 \end{pmatrix}.$$

The fundamental matrix is

$$\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{8} & \frac{1}{2} \end{pmatrix}^{-1} = \begin{pmatrix} \frac{8}{3} & \frac{8}{3} \\ \frac{2}{3} & \frac{8}{3} \end{pmatrix}.$$

We obtain again $\tau_0 = 8/3 + 8/3 = 16/3$ and $\tau_1 = 2/3 + 8/3 = 10/3$.

8. Let X_n be a Markov chain with state set $S = \{0, 1, 2, 3\}$ and transition probabilities:

$$p_{11} = p_{22} = p_{33} = \frac{1}{4};$$

$$p_{10} = p_{20} = p_{30} = \frac{1}{4}; \quad p_{01} = p_{02} = p_{03} = \frac{1}{3};$$

$$p_{12} = p_{13} = p_{21} = p_{23} = p_{31} = p_{32} = \frac{1}{4}.$$

- (a) Draw the transition diagram, write the transition probability matrix $\mathbf{P}$, and find the stationary distribution.
- (b) Let $T_0 = \min\{n \geq 1: X_n = 0\}$. Using first step analysis, compute the expected times $m_{i0} = \mathbb{E}(T_0 | X_0 = i)$ for each $i \in S$. Apply the fundamental matrix of the chain to verify the results obtained for each m_{i0} . Provide an explanation for the value of m_{00} .

Solution

- (a) The transition probability matrix is

$$\mathbf{P} = \begin{pmatrix} 0 & 1/3 & 1/3 & 1/3 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{pmatrix}.$$

The chain is finite and irreducible. Concerning its stationary distribution, we can take advantage of the symmetry of the chain. This implies that $\pi_1 = \pi_2 = \pi_3$. Then, the balance equation, together with the normalization condition, are:

$$\pi_0 = \pi_1 \cdot \frac{1}{4} + \pi_2 \cdot \frac{1}{4} + \pi_3 \cdot \frac{1}{4} = 3\pi_1 \cdot \frac{1}{4},$$

$$\pi_0 + \pi_1 + \pi_2 + \pi_3 = \pi_0 + 3\pi_1 = 1.$$

Solving the system we get $\pi_0 = 1/5$ and $\pi_1 = \pi_2 = \pi_3 = 4/15$.

- (b) Using again the symmetry of the chain, it must be $m_{10} = m_{20} = m_{30}$. Then, by using first step analysis, we have

$$\begin{aligned} m_{00} &= \frac{1}{3}(m_{10} + 1) + \frac{1}{3}(m_{20} + 1) + \frac{1}{3}(m_{30} + 1) = m_{10} + 1, \\ m_{10} &= \frac{1}{4} + \frac{1}{4}(m_{10} + 1) + \frac{1}{4}(m_{20} + 1) + \frac{1}{4}(m_{30} + 1) \\ &= \frac{1}{4} + \frac{3}{4}(m_{10} + 1) = 1 + \frac{3}{4}m_{10}. \end{aligned}$$

Solving the system, we get $m_{00} = 5$, and $m_{10} = m_{20} = m_{30} = 4$.

The fundamental matrix of a finite and irreducible Markov chain is $\mathbf{Z} = (\mathbf{I} - \mathbf{P} + \mathbf{\Pi})^{-1}$, where $\mathbf{I}$ is the identity matrix, $\mathbf{P}$ is the transition probability matrix, and $\mathbf{\Pi}$ is the square matrix with all rows equal to the stationary distribution vector

$$\boldsymbol{\pi} = \left(\frac{1}{5}, \frac{4}{15}, \frac{4}{15}, \frac{4}{15} \right).$$

Calculating this inverse we get

$$\mathbf{Z} = \frac{1}{75} \begin{pmatrix} 63 & 4 & 4 & 4 \\ 3 & 74 & -1 & -1 \\ 3 & -1 & 74 & -1 \\ 3 & -1 & -1 & 74 \end{pmatrix}.$$

Using the fundamental matrix $\mathbf{Z}$, we find that

$$m_{10} = m_{20} = m_{30} = \frac{z_{00} - z_{10}}{\pi_0} = \frac{63/75 - 3/75}{3/15} = 4,$$

as previously stated. However, if we attempt to calculate m_{00} using the formula

$$\frac{z_{00} - z_{00}}{\pi_0}$$

we get 0 because, when using $\mathbf{Z}$, the initial state counts.

9. Consider a game in which a coin with probability $\mathbb{P}(\text{heads}) = 2/3$ is tossed repeatedly. Denote the outcomes “heads” and “tails” by H and T, respectively. The game terminates upon the first occurrence of either the pattern HH or HTH, observed in two or three consecutive tosses, respectively.

- Define an absorbing Markov chain with state space $\{\emptyset, \text{H}, \text{HT}, \text{HH}, \text{HTH}\}$ to model this game. Classify each state and provide the corresponding transition probability matrix.
- Calculate the expected number of coin tosses required for the game to conclude.
- Given that the most recent toss resulted in “heads”, determine the expected number of remaining tosses until the game concludes.
- Determine the probability that the game concludes with the pattern HH and the probability that it concludes with the pattern HTH.

Solution

- Let $S = \{\emptyset, \text{H}, \text{HT}, \text{HH}, \text{HTH}\}$ (with the states in this order), where:
 - $\emptyset$: the game has not started.
 - H: the last performed toss landed “heads”.
 - HT: in the last two performed tosses the pattern “heads, tails” has been observed.
 - HH: in the last two performed tosses the pattern “heads, heads” has been observed, and the game finishes.

- HTH: in the last three tosses the pattern “heads, tails, heads” has been observed, and the game finishes.

The transition probability matrix $\mathbf{P}$ of this absorbing Markov chain is (the rows and columns are indexed by the states ordered as above):

$$\mathbf{P} = \begin{pmatrix} 1/3 & 2/3 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 2/3 & 0 \\ 1/3 & 0 & 0 & 0 & 2/3 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

The states HH and HTH are absorbing, and $\emptyset$, H, and HT are transient. Notice that $\mathbf{P}$ is already in canonical form.

- (b) Let us compute the fundamental matrix $\mathbf{N} = (\mathbf{I} - \mathbf{Q})^{-1}$:

$$\mathbf{N} = \begin{pmatrix} 2/3 & -2/3 & 0 \\ 0 & 1 & -1/3 \\ -1/3 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 27/16 & 9/8 & 3/8 \\ 3/16 & 9/8 & 3/8 \\ 9/16 & 3/8 & 9/8 \end{pmatrix}.$$

The expected number of tosses performed in the game is the expected time to absorption starting at $\emptyset$, $\tau_{\emptyset}$. Adding the entries of the first row of $\mathbf{N}$ we obtain

$$\tau_{\emptyset} = \frac{51}{16}. \quad (3)$$

- (c) If in the last performed toss the outcome is “heads”, the expected number of remaining tosses in the game is the expected time to absorption starting at state H, τ_{H} . This is the sum of the entries of the second row of $\mathbf{N}$,

$$\tau_{\text{H}} = \frac{27}{16}. \quad (4)$$

We can also find the expected times to absorption using a first step analysis. We have

$$\begin{aligned} \tau_{\emptyset} &= \frac{2}{3}(\tau_{\text{H}} + 1) + \frac{1}{3}(\tau_{\emptyset} + 1) = 1 + \frac{2}{3}\tau_{\text{H}} + \frac{1}{3}\tau_{\emptyset} \\ \tau_{\text{H}} &= \frac{2}{3} + \frac{1}{3}(\tau_{\text{HT}} + 1) = 1 + \frac{1}{3}\tau_{\text{HT}} \\ \tau_{\text{HT}} &= \frac{2}{3} + \frac{1}{3}(\tau_{\emptyset} + 1) = 1 + \frac{1}{3}\tau_{\emptyset} \end{aligned}$$

The solution of this linear system is

$$\tau_{\emptyset} = \frac{51}{16}, \quad \tau_{\text{H}} = \frac{27}{16}, \quad \tau_{\text{HT}} = \frac{33}{16},$$

in accordance with (3).

- (d) The absorption probabilities are given by

$$\mathbf{B} = \mathbf{N} \cdot \mathbf{R} = \begin{pmatrix} 27/16 & 9/8 & 3/8 \\ 3/16 & 9/8 & 3/8 \\ 9/16 & 3/8 & 9/8 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 2/3 & 0 \\ 0 & 2/3 \end{pmatrix} = \begin{pmatrix} 3/4 & 1/4 \\ 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}.$$

The probabilities $p_{\emptyset}$ and $p'_{\emptyset}$ that the game finishes by observing HH or by observing HTH are given by the two entries of the first row of $\mathbf{B}$, respectively. That is,

$$p_{\emptyset} = \frac{3}{4}, \quad p'_{\emptyset} = 1 - p_{\emptyset} = \frac{1}{4}.$$

Also, if $p_{\emptyset}$, p_{H} , and p_{HT} are the probabilities of finishing the game at the state HH, given that we start at the state indicated by the subindex, then

$$\begin{aligned} p_{\emptyset} &= \frac{2}{3}p_{\text{H}} + \frac{1}{3}p_{\emptyset} \\ p_{\text{H}} &= \frac{2}{3} + \frac{1}{3}p_{\text{HT}} \\ p_{\text{HT}} &= \frac{1}{3}p_{\emptyset} \end{aligned}$$

which solution is

$$p_{\emptyset} = \frac{3}{4}, \quad p_{\text{H}} = \frac{3}{4}, \quad p_{\text{HT}} = \frac{1}{4}. \quad (5)$$

The probabilities of finishing the game at the state HH , given that we start at the state indicated by the subindex, are the complementaries of (5):

$$p'_{\emptyset} = \frac{1}{4}, \quad p'_{\text{H}} = \frac{1}{4}, \quad p'_{\text{HT}} = \frac{3}{4}.$$

- 10.** A vending machine dispenses items for one euro and is equipped with a box, denoted B , with a capacity of $2k$ one-euro coins. Initially, box B contains k coins.

Each customer independently pays with a one-euro coin with probability p (where $0 < p < 1$), depositing the coin into box B , or with a two-euro coin with probability $q = 1 - p$, depositing the coin into a separate box B' . In the latter case, the machine dispenses one euro from box B as change. When box B becomes empty, an employee refills it with k one-euro coins. When box B becomes full, the employee removes k one-euro coins.

Let X_n denote the number of coins in box B after n individuals (including both customers and employees) have interacted with the machine.

- (a) Give the transition probabilities of the Markov chain $\{X_n : n \geq 0\}$ and classify its states.
- (b) Let $k = 2$. Determine the stationary distribution. What is the expected number of individuals who have interacted with the machine until box B again contains two coins? Compute $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k)$ if this limit exists, and justify the answer.
- (c) Let $p = 1/2$. Determine the expected number of customers served before the machine becomes empty or full for the first time.
[Hint: Apply first-step analysis to verify that the service time, starting with i coins in B , is $i(2k - i)$.]

Solution

- (a) Clearly X_n depends only of X_{n-1} so that the sequence of random variables $\{X_n : n \geq 0\}$ is a Markov chain. The variables X_n take values in $\{0, 1, \dots, 2k\}$, where i denotes the state in which the box B contains i coins. The transition probabilities are

$$p_{i,j} = \begin{cases} p, & j = i + 1, \quad 0 < j < 2k \\ q, & j = i - 1, \quad 0 < j < 2k \\ 1, & i \in \{0, 2k\}, \quad j = k \\ 0, & \text{otherwise} \end{cases}$$

We have a finite irreducible Markov chain. Thus, all the states are non-null recurrent.

- (b) If $k = 2$ the state sapce is $\{0, 1, 2, 3, 4\}$ and the transition probability matrix $\mathbf{P}$ is

$$\mathbf{P} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ q & 0 & p & 0 & 0 \\ 0 & q & 0 & p & 0 \\ 0 & 0 & q & 0 & p \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

The balance equations are

$$\begin{aligned} \pi_0 &= q\pi_1 \\ \pi_1 &= q\pi_2 \\ \pi_2 &= \pi_0 + p\pi_1 + q\pi_3 + \pi_4 \\ \pi_3 &= p\pi_2 \\ \pi_4 &= p\pi_3 \end{aligned}$$

Let $\pi_2 = \alpha$. Then, the stationary probability distribution vector is of the form $\boldsymbol{\pi} = \alpha (q^2, q, 1, p, p^2)$. Normalizing so that $\sum_i \pi_i = 1$ we get $\alpha = 1/(q^2 + q + 1 + p + p^2) = 1/(2 + q^2 + p^2) = 1/(3 - 2pq)$. Hence,

$$\boldsymbol{\pi} = \left(\frac{q^2}{3 - 2pq}, \frac{q}{3 - 2pq}, \frac{1}{3 - 2pq}, \frac{p}{3 - 2pq}, \frac{p^2}{3 - 2pq} \right).$$

The expected number of persons that have come to the machine until B contains again 2 coins is the mean recurrence time of state 2, which is $1/\pi_2 = 3 - 2pq$.

The period of, for instance, state 2 is one because starting from this state it is possible to return to it in two ($2 \rightarrow 1 \rightarrow 2$) or in three steps ($2 \rightarrow 1 \rightarrow 0 \rightarrow 2$). Hence, the chain is aperiodic and $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k) = \pi_k$, $0 \leq k \leq 4$.

- (c) Let T_i be the time until the machine is empty or full for the first time when the box B initially has i coins, and let Z be the indicator random variable that the first customer uses a one-euro coin. We have, for $0 < i < 2k$,

$$\mathbb{E}(T_i) = \mathbb{E}(T_i | Z = 1) \mathbb{P}(Z = 1) + \mathbb{E}(T_i | Z = 0) \mathbb{P}(Z = 0) = \frac{1}{2} (\mathbb{E}(T_{i+1}) + 1 + \mathbb{E}(T_{i-1}) + 1).$$

For $i = 0$ and $i = 2k$ we certainly have $\mathbb{E}(T_0) = \mathbb{E}(T_{2k}) = 0$.

Denote by $\tau_i = \mathbb{E}(T_i)$ so that, from $\tau_i = (\tau_{i-1} + \tau_{i+1})/2 + 1$ we get $\tau_{i+1} = 2\tau_i - \tau_{i-1} - 2$. Starting from $i = 1$,

$$\tau_2 = 2\tau_1 - 2, \quad \tau_3 = 2\tau_2 - \tau_1 - 2 = 3\tau_1 - 6,$$

and recursively, for $m \leq 2k$,

$$\tau_m = m\tau_1 - m(m-1).$$

By using $\tau_{2k} = 0$ we get $\tau_1 = 2k - 1$ and $\tau_i = i(2k - 1) - i(i - 1) = i(2k - i)$. In particular,

$$\tau_k = k(2k - k) = k^2.$$